When Schema Lexicons Fail: Diagnosing and Repairing Entity-Layer Collapse in a Production Knowledge Base

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Abstract

Background. In dictionary-driven entity enrichment the schema lexicon is effectively the extraction model. A category-word lexicon silently collapsed the entity layer of a 61,137-document scholarly knowledge base: 575 type-level entities ("method", "energy") carrying ∼19.2M links while every job-level monitor stayed green. **Method.** We diagnose the failure across extraction mechanism, lexicon provenance, and monitoring blindness; repair it with a corpus-attested lexicon built by LLM classification of corpus candidates under presence gating, deployed as a three-source recipe (3,702 terms); rebuild the entity layer under production load; and derive gold-free monitors led by a link-concentration diagnostic invariant to uniform rescaling of link counts. **Results.** A fixed-code, fixed-corpus experiment that varies only the lexicon reproduces the collapse signature alone (109±5 entities, top-10 concentration 0.513±0.016) and isolates the repair (692±27, 0.300±0.008; ten resamples). The rebuilt layer carries 1,871 entities and 25.1M links; distinct concepts per document rise 15.7→37.8 with annotated-document coverage 51.5%→98.5%; a risk-stratified 300-link audit reads 0.84 overall (LLM-assisted; no human gold yet). Matched spans are equally "valid" under old and new dictionaries (0.94 vs. 0.93, model-judged)—the collapse is a vocabulary-and-mass failure invisible to per-link audits—and no single gold-free monitor suffices (a composite four-leg rule reaches zero misses and zero false alarms on its own in-sample degradation archive). A three-domain replay on public corpora shows the failure form and the repair travel: batch-schema lexicons reproduce the collapse signature out of domain (C_{10} up to 0.899, silver recall

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≤0.009). **Limitations.** The generic tail survives repair (9% of links sit on the symbol "be"); retrieval-facing gains are not attributable to the lexicon alone; annotation is model-based (all judged rates are Qwen-family readings; cross-family arbitration touches 8–29% of rows), bounded by single-author human verification of a 197-term stratified sample (confirm 0.934)—an independent dual-annotator human gold is the declared next step; alarm bands are post-hoc candidate thresholds.

Keywords: knowledge base construction, schema quality, terminology extraction, entity linking, production incident, observability

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1. Introduction

Large scholarly knowledge platforms increasingly generate their subject schemas with LLM assistance: a batch tool drafts entity types with a handful of aliases each, a validator checks structural well-formedness, and the schema is bound to knowledge bases as configuration (our own batch generator is the primary failure source in this account; LLM-for-ontology generation is surveyed in [15]). This is attractive at scale—our deployment covers 80,855 parsed journal articles (80,852 active, 7,229,225 text chunks) across two large knowledge bases and several smaller ones—and it keeps humans out of a loop that would otherwise require domain lexicographers per discipline.

This paper is about what that design decision does to the entity layer when the enrichment pipeline extracts mentions by dictionary scan. In our system, stage-K1 mention extraction is deliberately LLM-free: a schema's `entity_types` main names and aliases constitute the scan dictionary. The choice is defensible—dictionary scan is cheap enough to run over 7.2 million chunks, deterministic, idempotent under re-ingest—and it performed adequately on the knowledge base whose schema had been hand-evolved over years. It failed catastrophically, and silently, on the two knowledge bases whose schemas came out of the batch generator.

The failure signature: a 61,137-document nuclear-electronics knowledge base (104112E6) accumulated **19.2 million entity links onto 575 canonical entities**; the entities were not domain concepts but type words—"method", "parameter", "energy". A 13,408-document physics knowledge base (693372D5) showed the same failure pattern at 2,018 entities. Meanwhile every operational signal was green: enrichment jobs completed normally, link volume read as growth, and retrieval endpoints

answered. The entity graph had degenerated into a bipartite "type-to-document" index while presenting, to every surface-level check, as a healthy knowledge graph.

We contribute:

1. **Incident forensics.** A forensic record of production entity collapse induced by schema-lexicon quality (Section 3)—575 type-level entities carrying ∼19.2M links under green job monitors—together with the full-chain rebuild under live load (Section 5: 766,501 re-enqueued jobs; 575→1,871 entities, ∼19.2M→25.1M links; a window-level 137× link-throughput recovery once the repair's own lexicon exposed a latent $O(N^2)$ alias check) and an incident log whose own corrections (a silent INSERT-IGNORE mis-write that dashboards missed; a logged "zero-growth" window contradicted by row timestamps) argue that dashboard-derived narratives themselves require row-level verification.
2. **Single-variable causal isolation of the lexicon.** With pipeline code and corpus both fixed, swapping the dictionary alone reproduces the collapse signature and its repair across ten random resamples of one library (Section 6), and the same swap replays out of domain (Section 8): the lexicon, not the pipeline, is the causal variable. The deployed remedy—a three-source corpus-attested recipe (3,702 terms) whose decisive mechanism the cross-domain ablation attributes to corpus-candidate classification under presence gating—is documented as the treatment under study; we claim the diagnosis and the isolation, not a new term-extraction method.
3. **Gold-free monitor calibration, led by top-10 link concentration.** The scale-invariant concentration diagnostic C_{10}, a top-entity name audit, and a composite four-leg rule, calibrated on 13 production libraries and verified in-sample on a graded-degradation archive (Sections 6–7), with blind spots stated per detection target and thresholds offered as deployment-derived triage bands, not classifiers.

The larger claim we defend: in dictionary-driven pipelines, **the schema lexicon is not descriptive metadata but operative extraction surface**. Treating it as configuration-file decoration—easily batch-generated, rarely audited—is a structural mistake that this incident renders measurable.

2. Background: dictionary-driven enrichment

Our enrichment pipeline (four stages per document: mentions → links → relations → graph) extracts mentions in stage K1 by normalized dictionary scan: every schema alias match over chunk text yields a `raw_mention` with an entity-type hint; stage K2 links mentions to canonical entities by normalized form + type; K3 lifts selected mention pairs into typed relations with an LLM pass; K4 projects the per-document entity subgraph into the graph store and search-side entity index. Retrieval consumes all four: an entity channel seeds personalized PageRank, a co-occurrence channel, and entity-vector similarity all trace back to linked mentions.

Why no LLM in K1? Cost and determinism. At 7.2M chunks, a per-chunk LLM extraction pass is a material recurring expense and a re-ingest hazard (non-determinism breaks idempotent rebuild); a scan is $O(\text{corpus})$ with zero marginal model cost. The design bets everything on one input: the dictionary. That bet is invisible in the architecture diagram and total in its consequences—a point Section 3 makes empirical.

The healthy control in our own deployment: a hydrogeology knowledge base whose schema was hand-evolved to 3,742 genuine domain terms (78 documents → 1,608 canonical entities, well-distributed). The contrast cases: schemas produced by our batch tool, whose output per entity type is a small set of *category* aliases (4–8 per type)—a taxonomy of the subject, not a terminology of it (the distinction between terminological and taxonomic vocabulary is classical [16], and the adjacent question of how to measure knowledge-graph quality is itself an established literature [10]).

Neither half of our pipeline is novel in isolation. Dictionary- or gazetteer-driven entity recognition is one of the oldest NER traditions [6], and corpus-statistical term recognition with linguistic evidence goes back to C-value/NC-value [5]—our corpus-candidate classification source (Section 4) is that automatic-term-recognition program reborn with an LLM in the classifier seat, and recent work extracts terminology directly with LLM-plus-retrieval pipelines [4]. The noise side is equally classical: gazetteers were known early as both the strength and the failure surface of dictionary NER [7], UMLS-scale dictionary lookup underlies the canonical biomedical recognition system MetaMap [8], and distant supervision—labels projected through dictionaries—prices the same false-positive mass as its founding cost [9]. Our incident is that literature's production-KB echo at a new layer: the gazetteer is not a curated resource that degrades, it is batch-generated

schema content, and its noise is invisible per-link while fatal in the aggregate. Automatic schema and KG construction has meanwhile advanced to dynamic schema induction over corpora (AutoSchemaKG [1]), incremental schema evolution (DIAL-KG [2]), and joint evaluation of extraction quality, schema adherence, and stability [3]. Our subject is complementary and upstream of a different risk: in *dictionary-scan deployment*, where the schema's alias list is the extraction surface, we show that schema legality and even per-link mention validity can coexist with asset-level semantic collapse—a failure locus those methods do not measure, and one our diagnostics target directly; we claim no better schema-induction or term-extraction algorithm. On the term side, evaluation traditions for automatic term extraction are annotation-guided and domain-stratified (TermEval/ACTER across domains and languages [13]); our silver/dual-model references follow that spirit at zero marginal human cost, with the same caveats about instrument dependence. Diachronic analyses show domain vocabularies drift measurably even without failure [14]; our monitors treat lexical drift as one more reason lexicon content, not just schema shape, needs standing statistics. Likewise, detecting and repairing degraded knowledge-graph content is an established concern [10], and production data-validation systems routinely gate model inputs with schema- and statistics-level checks [11, 12]. What this incident supplies is a specific failure locus: when the dictionary *is* the extraction model, a structurally valid but terminologically hollow schema passed every gate *this pipeline had in place*—schema-shape validation, ingestion checks, job-level monitoring. That is a statement about our gates, not about the prior art: distribution-statistics monitoring is the right tradition, but the statistic must be aimed at the lexicon's terminological content—none of our gates asked whether a lexicon's entries are domain-specific terms, the question term-recognition research takes as its starting assumption. The monitors we propose (Section 7) target that acceptance gap.

3. The collapse, forensically

The observation. During an unrelated storage investigation (2026-10-08), a routine entity-quality sample on 104112E6 returned type words where domain concepts were expected. The full picture (Table 1):

61,137 documents carrying ∼19.2M links is ∼315 links per document on average, and ∼33,000 links per entity: every document "mentions" the type

Table 1: The collapse, measured. "bckp." = 2026-10-06 cold-backup snapshot (mid-enrichment); "disc." = discovery-day counts; the hydraulic reference case is measured 2026-10-10.

KB	Docs	Entities	Links	Schema provenance
104112E6 (nuclear)	61,137	**575** disc. / 503 bckp.	∼**19.2M** disc. / 4.81M bckp.	batch v2.0
693372D5 (physics)	13,408	**2,018**	2.23M (bckp.)	batch
hydrogeology (ref.)	78	1,608	719,515	hand-evolved

vocabulary constantly. The graph is a bipartite type–document index that is row-level indistinguishable from a populated knowledge graph.

The incident timeline (dates stated plainly, because a repair of this scale compressed into a two-day window is exactly the kind of claim a reader should be invited to check):

The compression is real and we do not smooth it over: the tooling was written under incident pressure on day one, and the evaluation experiments are retrospective audits of a settled production state, not a pre-registered protocol—a limitation Section 10 keeps visible.

Layer 1 — mechanism. K1 scans for schema aliases. When the alias set for type `method` contains common words ("method", "approach", and their CJK equivalents), every occurrence of those words in 7.2M chunks becomes a mention, and K2's normalized-form+type linkage folds them onto one canonical entity per (form, type) pair. The mechanism executed faithfully on degenerate input.

Layer 2 — provenance. The dictionaries were the schemas. The batch generator, prompted to "draft entity types with aliases for subject X", produced exactly that: type-level category vocabulary. Nothing in its validator checks whether aliases are *domain-specific*—structural validity is the gate, and category words are structurally valid.

Layer 3 — the reference case. Hydrogeology's 3,742 hand-evolved terms over 78 documents yield 1,608 entities. We call it a reference case, not a controlled comparison: domain, scale, register, and lexicon provenance all differ, so it establishes plausibility, not causation — which is what the isolated experiment of Section 6 (6.5) is for. Per-document link mass differs by ∼22×

Table 2: Incident timeline (events compressed; full narrative in the text).

Date	Phase	Event
10-08	discover	storage investigation surfaces type-word entities
10-08	root-cause	three-layer diagnosis; lexicon tool built, run, committed (v2.1)
10-08	rebuild	teardown + 766,501-event re-enqueue; B95 silent misattribution window
10-09	drain	four stages complete; stats recomputed; vectors reprojected
10-09	extend	two smaller libraries with the same failure pattern treated (same playbook)
10-09/10	audit	scoped refresh, channel ablation, link audit, concentration fleet-wide
10-09/10	manuscript	drafted, revised; retrospective — not pre-registered

(control ∼9,200 links/document vs. 315 collapsed / 410 repaired on KB1), but that gap is a document-length artifact, not a linking-rate difference: the control is textbook-scale material at ∼1,913 chunks per document against KB1's 99, and the per-chunk link rates are comparable (4.8 vs. 4.1).

Why nothing fired. Job-level observability was all green (jobs DONE, throughput nominal, error rate zero). Link volume—if anything—read as *success* (19.2M links!). The observables that expose the collapse are (a) the concentration of links on the top entities—the collapsed state measures 0.419 with its top-10 entities being pure type vocabulary, versus 0.18 for the healthy control (Section 7's monitor (a) gives the measurement and its provenance)—and (b) the name distribution of top-linked entities, which is pure type vocabulary. Entity cardinality per document—a factor of 2,192 below the reference case (0.0094 vs 20.6), over three orders of magnitude—corroborates, but cardinality is a coarse signal gameable in both directions (the small-library treatment of Section 10 shows category-word aliases inflating it), which is why it is the secondary, not the primary, monitor. Neither observable is provided by standard pipeline dashboards by default. This blindness generalizes to any dictionary-driven system whose dashboards count jobs and rows.

category-word lexicon (v2.0)	**repaired lexicon (v2.1)**
`method` → type *method*	`hpge detector` → *semiconductor_detector*
`energy` → type *energy*	`preamplifier` → *circuit*
`time` → *physical_quantity*	`neutron activation analysis` → *method*
575 entities / 19.2M links	1,871 entities / 25.1M links
48.5% of documents unannotated	98.5% of documents annotated

Figure 1: Mechanism: the schema lexicon *is* the extraction model. Identical scan-and-link code over the same corpus; only the dictionary differs (left column illustrative; right column verbatim (surface, type) pairs from v2.1).

4. Corpus-Attested Lexicon Construction and Validation

Figure 2 shows the three-source admission process and its quality boundary.

The repair requirement: produce genuine domain terminology at scale, without per-subject lexicographers, without hallucinated terms—and make it a repeatable tool, not a one-off patch. Our generator implements three sources, all gated:

1. **Seed transplant (2,284 terms).** A healthy near-domain library (the RADAI corpus knowledge base, same academic register) donates its terms with types. Transplant is the highest-precision source; corpus attestation below still applies.
2. **Corpus-candidate classification (7,391 candidates).** High-frequency n-grams from the target corpus's own text are proposed—concretely, the top 8,000 n-grams (unigrams and bigrams) by chunk frequency over a 3,000-chunk-per-KB sample, keeping candidates with frequency $\geq$5 after a boilerplate and stop-word filter (section vocabulary such as *acknowledgements/approach/using*; closed-class English words). An LLM classifies each surviving candidate into a schema entity type or rejects it. The corpus floor is a corpus-presence gate—attestation means the term occurs in the corpus, which is necessary but not sufficient for termhood (the "be" case of the name-quality section is exactly this gap); it prevents hallucinated entries, not generic-word admissions.

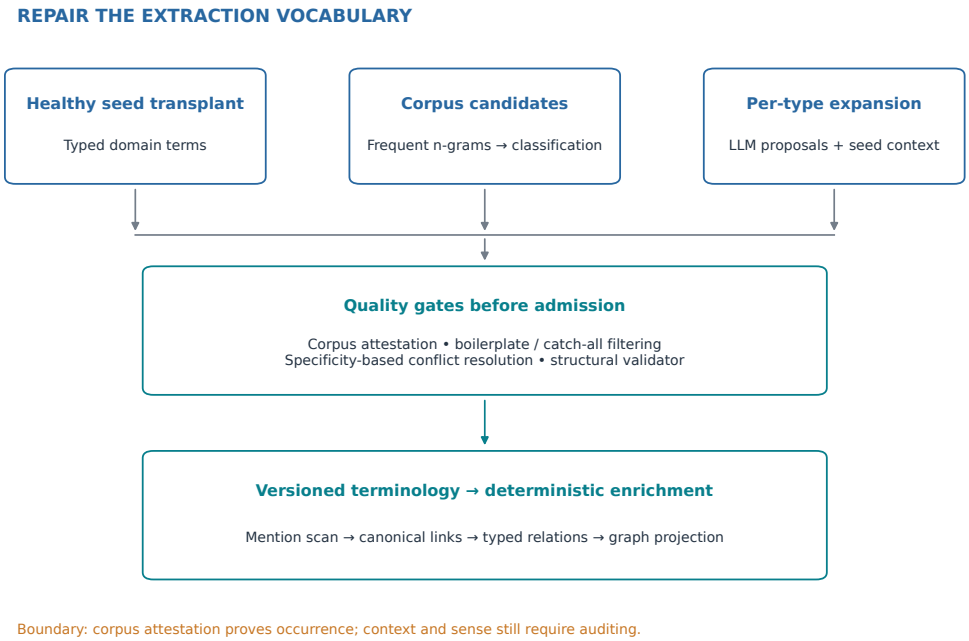


Figure 2: Three-source terminology generation and deterministic enrichment. Healthy seeds, classified corpus candidates and per-type expansion proposals pass corpus-attestation, vocabulary, conflict-resolution and structural gates before entering a versioned schema. The admitted vocabulary drives mention scanning, canonical linking, typed-relation adjudication and graph projection. Attestation verifies occurrence; contextual sense and domain specificity still require an independent link-quality audit.

3. **Per-type expansion ($\leq$60 per type).** For each entity type the LLM proposes additional domain terms, given the type description and the transplanted seeds as in-context anchors; the same corpus-attestation gate applies. The bound keeps any single type from flooding.

Generation parameters, for reproducibility: the classifier/expander is an AWQ 4-bit build of the Qwen3.6-35B-A3B checkpoint (community build `cyankiwi/Qwen3.6-35B-A3B-AWQ-4bit`; we retain the deployment string verbatim since the endpoint serves it under a historical alias), served via vLLM behind a round-robin router, temperature 0.2, $\leq$4,000 output tokens per call, three retries on empty or non-JSON responses; the entire three-source run for both schemas is a single idempotent invocation of the generator.

Quality gates beyond attestation: *catch-all types* (`term`, `term_cn`, `model_or_code`, `concept`, `question`) accept no classified noise—they are the natural sink of classification uncertainty and the main entry route for taxonomy garbage; a boilerplate filter removes layout/section vocabulary; cross-type alias conflicts resolve by a specificity ordering (the more specific type wins); and the schema validator runs as a final structural gate before any version bump. The same pass repaired 223 legacy cross-type conflicts in existing schemas.

Output: **nuclear_electronics v2.1 = 3,702 terms; physics v1.1 = 1,279 terms**. 104112E6 rebinds to nuclear_electronics (its corpus is nuclear-domain journals). Both versions are idempotent re-runnable—the tool is now part of the schema lifecycle, not an incident artifact.

An honest note on the asymmetry (physics 1,279 versus nuclear 3,702): the physics corpus is smaller and its seed transplant thinner; v1.1 is a floor, not a ceiling, and the runbook treats small-physics libraries as follow-up work (Section 10).

5. Rebuild at scale

Repairing live knowledge bases requires tearing down the collapsed entity layer without stopping ingestion or retrieval. The rebuild driver executes per knowledge base: delete graph-store vertices with edges; delete the four relational tables (mentions, links, relations, graph-projection queue rows) in bounded chunks; delete the search-side entity projection by query; then re-enqueue all documents through the four stages—**766,501 enqueue events** across the two affected libraries. The count is events, not distinct doc-stages (74,545 documents $\times$ 4 stages = 298,180); Table 3 reconciles the surplus to the repair window's re-enqueue waves—the job table holds one row per doc-stage with an attempts counter, so every wave re-uses rows, and teardowns reset the counters, which is why only the logged resets are exact. The rebuild ran against the production fleet (five multi-worker enrich instances, one links-dedicated consumer per KB) while normal retrieval served.

The rebuild doubled as the system's first full-corpus stress test of the *rich-lexicon* configuration; four secondary engineering incidents surfaced and were fixed with measured effect (an unbounded projection join, two link-insertion deadlock classes, and an $O(N^2)$ alias check the richer lexicon itself amplified—fleet link throughput 8/min $\rightarrow$ 1,100/min, a window-level aggregate). Their per-pathology accounting is engineering, not evidence, and

Table 3: Reconciling the 766,501 enqueue events. Exact rows are row-countable; ≈ rows are operations-log wave estimates, and the final remainder is balanced rather than logged—teardowns reset the attempts counters, so exact per-wave attribution beyond the logged resets is unrecoverable.

Component	Events
initial re-enqueue, 74,545 docs × 4 stages	298,180
silent-empty links reset (KB1 51,459 + KB2 24,372)	75,831
small-library treatment (1,443 docs × 4)	5,772
relations/graph zero-effect reset, both libraries	≈149,090
watchdog graph-deletion recovery, KB1 rel.+graph	≈122,274
partial-wave re-dispatch, gated window (balancing)	≈115,354
total	766,501

moves to Appendix C; one reading matters here: the collapsed lexicon (575 de-facto terms) masked the $O(N^2)$ hot path because N was tiny—correcting the lexicon changed the pipeline's computational regime, so lexicon quality is load-bearing in the most literal sense.

The rebuild also supplied its own fifth pathology, and revisiting it with the registry's own timestamps forced a correction to our incident log that we report in full. The original record: a single-statement link writer shipped with a parameter-order bug (the SQL's first placeholder expected the knowledge-base id; the call bound a term form there), jobs completed green for hours, and per-KB link counts read zero growth until a row-count reconciliation caught it. What a minimal reproduction on the production engine (MySQL 8.4, `STRICT_TRANS_TABLES`, faithful DDL) now establishes is sharper than "rows were swallowed": under `INSERT IGNORE`, the NULL produced by `UNHEX('alloy')` is **replaced by the implicit default 0x00×16 and the row is inserted**—into a ghost bucket invisible to every per-KB query and monitor (warnings 1411 and 1048 fire; nothing fails)—and a later correct re-insert for the same (mention, entity) pair is **silently skipped by the unique key**, so the orphan actively poisons the repair unless deleted first. Production carries zero 0x00 rows today, so the recovery path included that cleanup. A second correction concerns duration: the operations log recorded "seven hours of zero growth", but the surviving rows' own `created_at` histogram (Figure 3) shows KB1 inserting at 1.0–2.3M rows/hour continuously through that nominal window and KB2's entire link layer landing in a single hour—the zero-growth reading

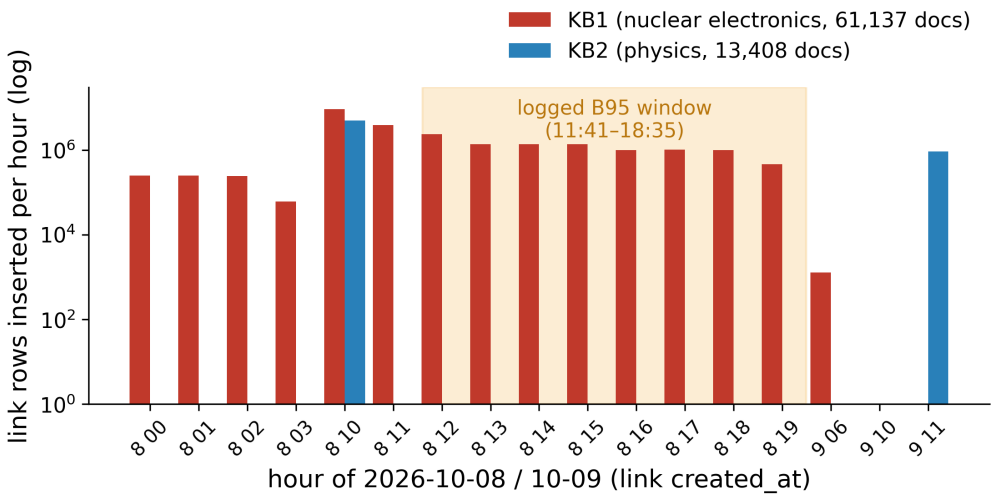


Figure 3: Rebuild telemetry from the link registry's own timestamps (hourly `created_at` counts). KB2's entire link layer lands in the 10:00 hour (5.0M rows); KB1 drains across the day at up to 9.2M rows/hour. The shaded band is the *logged* B95 window (11:41–18:35); inserts continue through it, which forces the timestamp correction reported in the text.

cannot be reconciled with the timestamps and most plausibly reflects a monitoring view scoped to the buggy writer's partition rather than the whole table. We keep both the original log entry and its correction in the record: incident narratives written from job dashboards are themselves unverified claims; row-level forensics is the only authoritative account.

Rebuild final state. All four stages drained across both knowledge bases: mentions, links, relations, graph projection. The repaired entity layer, measured: **KB1 1,871 entities carrying 25,097,704 links (from 575 / ∼19.2M collapsed), 44,043 relations with 1.56M evidence rows (from effectively zero); KB2 2,730 / 5,009,755 / 12,774.** Spot-checked canonical entities (germanium detector, hpge detector, neutron, preamplifier, scintillation detector) all project correctly into the graph store. Retrieval on KB1 answers in 0.6–1.3 s warm with all channels active.

6. Before/after

Figure 4 compares entity cardinalities across the two main and two auxiliary repaired libraries.

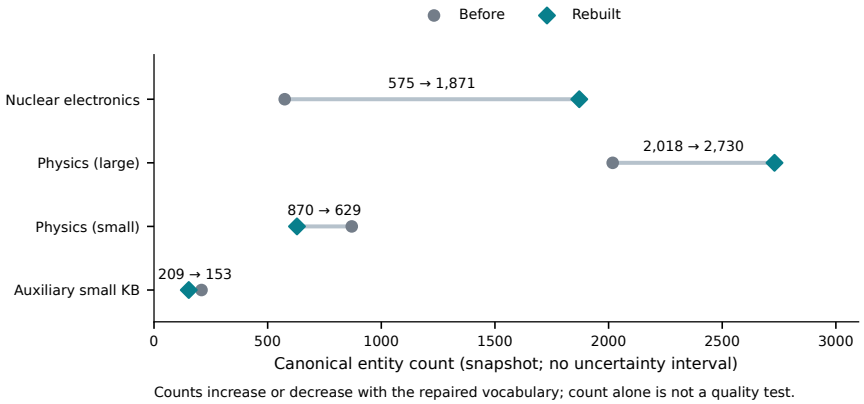


Figure 4: Entity counts before and after vocabulary repair. Gray circles are pre-repair snapshots; teal diamonds are rebuilt snapshots. Nuclear electronics and the large physics library expand, whereas the small physics library (D2B4AD67) and auxiliary library (F54CE546) contract. These are complete registry counts, not sample estimates, so no uncertainty intervals are shown. Increasing cardinality alone cannot establish improved link quality.

Entity layer, same-snapshot framing. The before/after rows mix two pre-repair snapshots; the like-for-like comparison is the 10-06 cold backup vs. post-repair (503→1,871 entities, 4.81M→25.1M links), while the discovery-day figures (575, ∼19.2M) record the incident at its worst, after mid-enrichment evolution continued past the backup. The headline repair, with snapshot provenance stated (collapsed-era link metrics from the 2026-10-06 cold backup, a mid-stream snapshot two days before discovery; terminal collapsed totals from the discovery-day count; repaired metrics 2026-10-10):

Concepts-per-document replaces the raw entity/document ratio as the cardinality-family metric: its numerator scales with documents, it is readable by domain experts, and the coverage column exposes the collapse's most user-visible symptom—half the library had *no* entity annotations at all. The concentration drop 0.419→0.345 is honest about its own limits: the pre-repair figure is a mid-stream snapshot (4.81M links; the terminal state was 19.2M on the same entities, which can only push concentration toward the type-word head), and part of the drop comes from the entity population growing—the name audit below remains the sharper discriminator. Absolute link counts *rise* after repair in both libraries (KB1 ∼19.2M→25.1M; KB2 2.23M→5.0M,

Table 4: Before/after, entity layer, with snapshot provenance.

Metric (KB1)	Collapsed	Repaired
canonical entities	575 (disc.) / 503 linked (10-06)	**1,871**
entity links	∼19.2M (disc.) / 4.81M (10-06)	**25,097,704**
typed relations	∼0 effective	**44,043** + 1.56M evid. rows
top-10 link concentration	0.419 (10-06)	**0.345**
concepts / document (all-doc basis)	15.7 (51.5% covered)	**37.8** (98.5% covered)
KB2: concepts / document	29.5 (91% covered)	**61.4** (91% covered)

same-snapshot backup on both sides): the repaired lexicons carry thousands of real terms that match more often. Count was never the quality signal—distribution (concentration) and names (top-entity audit) are.

Name quality, honestly. The top-linked entities after repair are `be` (2.26M links), `alpha`, `gamma`, `beta`, `method`—generic words and short ambiguous symbols still dominate the link head. The corpus-attestation gate can only guarantee that a term *occurs*; it cannot distinguish a high-frequency common word from a domain term. The `be` case is instructive because it is not even a ranking problem but a normalization one: the element symbol *Be* and the English auxiliary verb *be* are distinct in the source text, but K1's scan normalizes case (ASCII terms match case-insensitively with word boundaries), so every auxiliary "be" in 7.2M chunks links to the beryllium entity. A case-sensitive allowlist for short chemical symbols plus an English stop-word exclusion at admission would remove the bulk of this mass; `alpha`/`gamma`/`beta` are harder—genuine nuclear-spectrum symbols that also serve as generic Greek-letter variables—and need context-type constraints, not stop-word lists. Genuine terminology (hpge detector, neutron, preamplifier, …) is present and correctly linked and projected—but the repair relocated the lexicon-quality problem from "no terminology at all" to "terminology plus a generic tail." We consider this the honest headline of the incident: repairing a collapsed lexicon is one battle; lexicon hygiene is a permanent operational discipline (Section 7's monitors exist precisely for this).

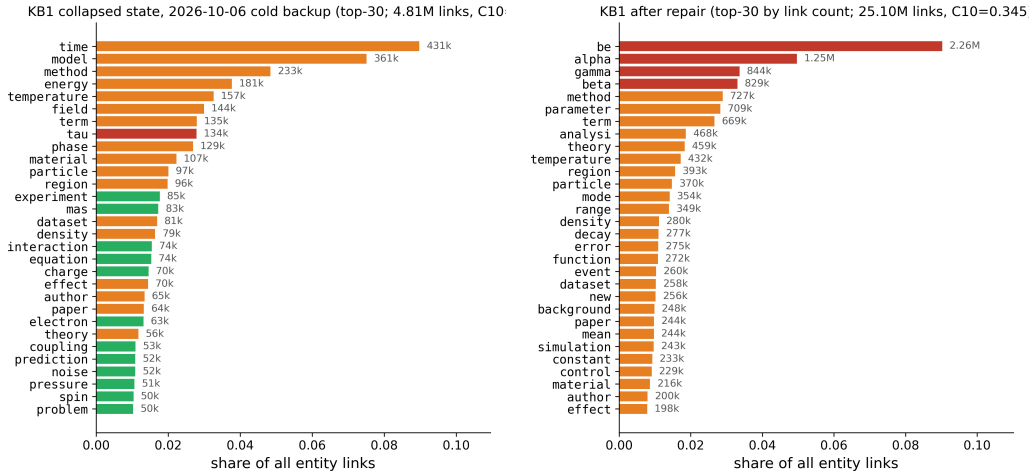


Figure 5: Top-30 entities by link share, KB1, before (2026-10-06 cold backup; $C_{10} = 0.419$) and after repair ($C_{10} = 0.345$). Categories follow a fixed three-way rubric (domain term / generic academic word / short ambiguous symbol), assigned by the authors per entity; the rubric and per-entity assignments ship with the artifact. The head is dominated by generic vocabulary in both states; repair diversifies the mid-tail (visible as the shorter bars) but does not dethrone it.

Retrieval trajectory, with a channel-level attribution bound. On the scoped evaluation slices (same runner, same corpus scope) the system moved from the collapsed-era run (2026-09-24) to the rebuilt state (2026-10-09): associative queries cov@10 0.533→**0.599** (+6.6pt, n=48), single-intent queries 0.603→**0.675** (+7.3pt, n=39) with MRR −2.0/+5.0pt respectively. The two-week window also carried retrieval-configuration evolution (document-fusion re-ranking, bilingual alias expansion, kNN search-depth tuning), so the trajectory measures system evolution *including* the lexicon repair. To bound the attribution we ran a single-variable channel ablation under the identical post-rebuild configuration (in-process planner, same corpus scope, same questions): a *no-entity* arm with every entity-dependent channel disabled (co-occurrence, graph, entity-vector, PPR) against the full configuration. Result: **@10 any-hit and document coverage are unchanged on both families** (nea 0.599/0.938, ner 0.670/0.923); the only movement anywhere is ner MRR +0.0006 (CI [0, 0.0018], ns) and nea cov@20 −1.0pt, 95% CI [−3.1, 0.0]—while p50 latency falls from 1.46 s to 0.52 s. The ablation bounds these channels' marginal contribution in this configuration, not

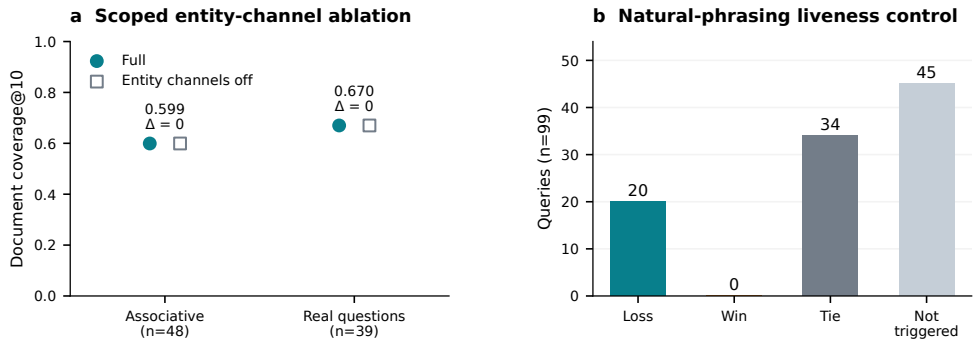


Figure 6: Distribution-dependent attribution of entity-channel effects. (a) Under the identical post-rebuild configuration, disabling all entity-dependent channels leaves cov@10 unchanged on the scoped associative ($n = 48$) and real-question ($n = 39$) sets. Points are observed set averages; no marginal confidence intervals were archived for these levels. (b) On the natural-phrasing liveness set ($n = 99$), co-occurrence fires on 54 queries: its removal causes 20 gold-hit losses, zero wins and 34 ties. The other 45 queries are shown explicitly as not triggered. The liveness experiment has a different question distribution and is not a before/after estimate of the lexicon repair's causal effect.

the semantic assets' value at large. The co-occurrence channel fired on only 6/48 association queries (its adaptive gate suppresses it when base retrieval already serves co-mention documents) and changed no outcome.

Figure 6 separates the scoped zero-effect finding from the natural-phrasing liveness control.

Channel-liveness check (the ablation's alternative explanation, closed). A null @10 result admits a defensible alternative reading—that the entity channels are not wired, or that their gates never open—so we ran a forced-trigger check on the 99-question natural-phrasing set (template words banned), the distribution the co-occurrence gate was designed for: there the channel fired on **54/99 queries** (versus 6/48 on the scoped template set), killing it changed the top-10 on 26 queries, and the paired gold-hit comparison reads **20 losses / 0 wins / 34 ties** for the off-arm—removing the channel never helps and hurts on a fifth of the set. The channel is alive, the gate opens exactly where designed, and the scoped zero-effect is the gate closing because base retrieval already serves those co-mention documents—not a dead wire.

Run-to-run noise floor. Two same-configuration scoped runs hours apart (the refresh run and the ablation baseline) differ on exactly one of 39 single-intent queries (cov@10 0.675 vs 0.670)—a measured 0.5pt noise floor from nondeterministic channel timing that bounds every within-window comparison in this paper.

Alias-expansion ablation. The one entity-adjacent mechanism that plausibly carried the trajectory—bilingual alias expansion feeding BM25—was ablated on/off under the final configuration: association cov@10 +1.5pt (paired bootstrap CI [−0.6, +3.9], ns) and single-intent +0.1pt, at a p50 latency cost of 1.6→3.5 s. On the benchmark sets the retrieval-window trajectory is therefore not attributable to the lexicon through this mechanism either; where alias expansion pays is query distributions rich in dictionary surfaces (the natural-phrasing set), which the benchmarks under-represent.

The attribution this forces is precise and we state it against interest: the +6.6/+7.3pt trajectory is **not** carried by entity channels casting fusion votes. The entity layer's retrieval-facing value on these benchmark families runs through the lexical channel's bilingual alias expansion (entity names as BM25 terms—a mechanism that needs a real terminology to expand, and which the collapsed lexicon could not feed) and through structural surfaces (reranker context, knowledge-chain evidence for generation) documented in the companion system manuscript [17]; at the fusion layer itself, entity channels are currently a latency cost with zero measured recall benefit on these sets. The lexicon repair was necessary for every entity-consuming surface; the scoped trajectory number alone cannot be credited to it.

Link precision audit (LLM-assisted). Distributional monitors say nothing about whether individual links are *correct*. We audited a seeded, reproducible uniform sample of 100 links from the rebuilt KB1 registry (`ORDER BY RAND(20261009)`): for each link the mention span was programmatically verified against the chunk text (span integrity 100/100—offsets are exact), and an LLM judge (Qwen3.6-35B-A3B, temperature 0) received the ±250-character context with the span marked and answered a strict two-part rubric: (i) does the span denote the same concept as the entity's canonical name (different sense, different concept, or non-term mention ⇒ invalid; debatable ⇒ invalid + borderline); (ii) is the canonical name a domain-specific technical concept or a generic academic word. Results: **overall precision 0.85** (Wilson 95% CI [0.77, 0.91]); the domain stratum is **25/25 = 1.00** (CI [0.87, 1.00]) and the generic stratum 60/75 = **0.80** (CI [0.70, 0.88]). Every adjudication, span, and reason string is released. The reading is the name-

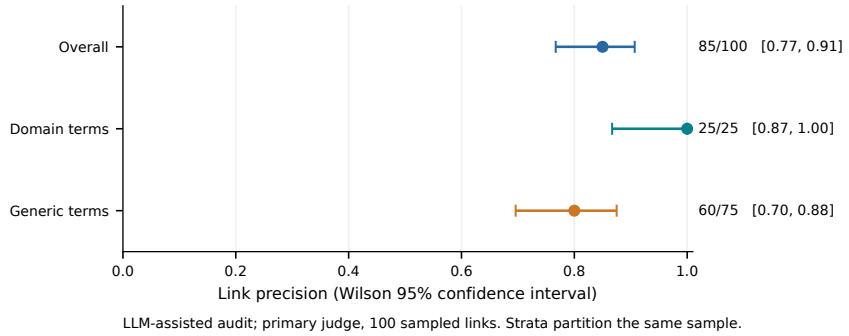


Figure 7: Link precision in the primary LLM-assisted audit of 100 sampled KB1 links. Points show valid-link fractions and horizontal intervals are Wilson 95% confidence intervals. The overall sample (85/100) comprises domain-term (25/25) and generic-term (60/75) strata; the three rows are not independent samples. Domain specificity separates the observed error mass, but the domain stratum is small and this audit is not human adjudication. Every sampled link is retained.

quality headline quantified: in this 25-link domain subsample no erroneous link was observed, and the residual error mass sits almost entirely in the generic tail—but "no errors observed at n=25" is not "error-free": the larger stratified audit's domain stratum reads 142/149, not 149/149. The audit is LLM-assisted rather than human-adjudicated—a limitation we keep visible (Section 10)—but its stratum split is exactly what the monitor design predicts, and the programmatic span check removes the cheapest class of audit noise.

Figure 7 displays the audit's uncertainty and stratum sizes.

Cross-model check (the judge-independence concern, addressed). The audit judge and the lexicon classifier are the same model family—a self-grading risk we take seriously. We re-adjudicated the identical 100 links with a second judge from a different model, checkpoint, and provider (Qwen3.8-Flash via an external gateway; temperature 0, identical rubric). Both judges are Qwen-series—vendor-level correlation is not excluded, and we label this a cross-model check, not an independent family confirmation. The second judge failed to emit parseable verdicts on 45/100—a flash-tier instruction-following failure we report as-is, not hide. The agreement figure therefore applies only to the 55-link parseable subset and cannot be read as confirmation of the full 100; on that subset, verdict

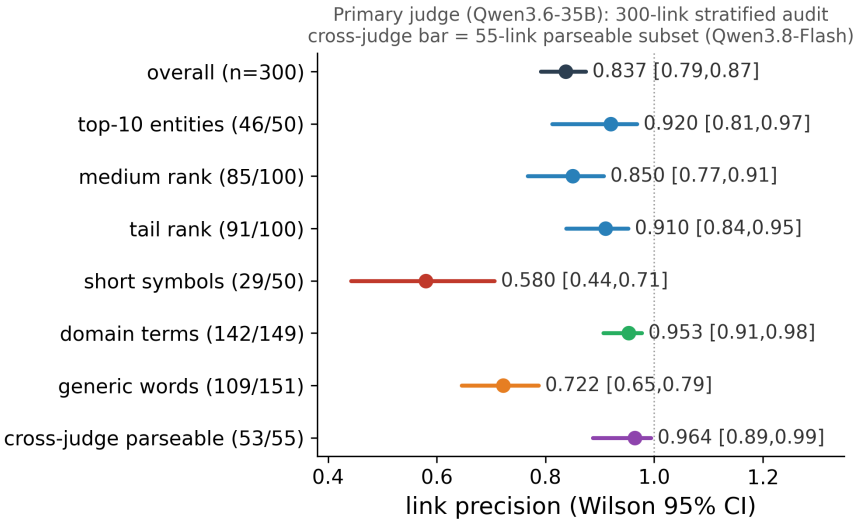


Figure 8: Link-precision with Wilson 95% CIs: 300-link stratified sample (primary judge); cross-judge bar applies to the 55-link parseable subset of the earlier uniform sample (45/100 unparseable); both judges are Qwen-series.

agreement with the primary judge is **53/55 = 96.4%** (50 both-valid, 3 both-invalid, 2 discordant—both discordances on the borderline generic word "term"), and the domain stratum agrees **14/14**. Where an independent model opined, it confirmed the primary adjudication; but vendor-level correlation is *not* confined to the unparseable subset—the 55 parsed links, including the domain stratum (n=25 primary, 14 cross-parsed), carry the same family risk. A human-adjudicated stratified sample remains the decisive control and is listed as such in Section 10.

Stratified 300-link audit (third review round). A uniform sample reads the head; the failure modes live in the tail. We re-audited with a stratified design: entities drawn uniformly within four strata (top-10 by link frequency; medium, ranks 11–100; tail, ranks 101+; short symbols, canonical names of $\leq$4 letters), then one link uniformly per entity—seeded, replayable, 50/100/100/50 links, same rubric and judge. Span integrity 300/300. The judged-valid rate for this sample is **0.837** (Wilson 95% CI [0.791, 0.874]; entity-cluster bootstrap over the 187 clusters, 10,000 resamples: [0.782, 0.887]); top-10 **0.920** [0.812, 0.968]; medium **0.850** [0.767, 0.907]; tail **0.910** [0.838, 0.952]; short-symbol **0.580** [0.442, 0.706]; across

the specificity axis, domain 142/149 = 0.953 and generic 109/151 = 0.722. The error mass is localized, not diffuse: the stratified design confirms the uniform reading at the head and pins the failure precisely on the short-ambiguous-symbol class—the more informative finding of the audit, and the motivation for the governance pricing below (not yet a demonstration of its post-treatment validity). Two scope notes: the 50/100/100/50 quotas target risk rather than production link frequencies, so the 0.837 figure is a stratum-mixed rate, not an unweighted production-wide link precision; and the 300 links come from 187 distinct entities (the head stratum's 50 links from just six; the short and head strata share "term"), so the intervals treat links, not concepts, as exchangeable.

The lexicon variable, isolated. The repair bundled lexicon, teardown, and fleet changes; this experiment isolates the lexicon alone. Design: a semantic replica of the K1 scan-and-link semantics (CJK substring match; ASCII word-boundary + optional plural, case-folded) and the K2 fold (normalized form + type), with **only the dictionary swapped**: v2.0 as committed at library creation (182 scan terms), v2.5 (mid-evolution, 2,932), v2.1 (the repair, 3,733 scan terms; three v2.1 counts appear in this paper and refer to different objects: 3,702 canonical schema terms (the committed lexicon), 3,733 scan terms after alias dedup (the replica frame), and 3,824 entries in the production composite dictionary (schema lexicon ∪ operational seed overlay)), plus two v2.1 governance variants (case-sensitive matching for short ASCII symbols; an English stop-word block at admission). Corpus frame: **ten seeded, uniformly random resamples of 2,000 chunks each** from the library's 5.83M visible chunks—independent draws from a *single* corpus, i.e. a sampling-stability check within one library; the sampling unit is the chunk, and in this corpus chunks map to documents almost one-to-one (sample 0: 2,000 chunks → 1,965 distinct documents, 1.02 chunks/doc), so "matched documents" below means matched chunks and no document is sampled twice evidence—cross-sample disjoint, deduplicated on redraw. The replica copies the scanner's matching and folding rules from the same source reading (CJK substring; ASCII word-boundary with optional plural, case-folded; fold key = normalized form + type); it is not instrumented against the live service path, so absolute levels may drift—the experiment's force is the within-frame arm contrast. A frozen-sample parity check bounds that drift: on 200 frozen KB1 chunks scanned with the *live* production dictionary (3,824 entries), every one of the 913 production mention spans is reproduced by the replica (span-level *superset*, not output equivalence: the per-term replica ad-

Table 5: Isolated lexicon comparison across ten independent corpus resamples (fixed code, fixed corpus, only the dictionary differs; mean±sd). Conc/doc is the mean number of distinct concepts per *matched* document (documents with ≥1 scan hit under that arm's dictionary); the arms' denominators differ by construction (text). The sd is within-corpus resample variance (ten draws from one library's chunks); it does not express cross-corpus or cross-library variation.

Arm	Links	Ents	C_{10}	Conc/doc	Gen.-head
v2.0 (collapse-era)	$2{,}969 \pm 140$	109 ± 5	$0.513 \pm .016$	$1.85 \pm .05$	$0.146 \pm .012$
v2.5 (mid-evolution)	$4{,}823 \pm 147$	421 ± 25	$0.322 \pm .009$	$2.51 \pm .05$	$0.090 \pm .006$
v2.1 (repair)	$10{,}042 \pm 251$	692 ± 27	$0.300 \pm .008$	$3.89 \pm .07$	$0.058 \pm .004$
v2.1 + case-sens. short	$8{,}758 \pm 228$	665 ± 28	$0.265 \pm .012$	$3.54 \pm .07$	$0.066 \pm .004$
v2.1 + stop-word	$9{,}226 \pm 251$	691 ± 27	$0.252 \pm .011$	$3.67 \pm .07$	$0.063 \pm .004$

ditionally reports 263 overlapping or nested occurrences that the production merged longest-first alternation suppresses by design, and a normalized-form bookkeeping rule differs for aliases that themselves end in "s"). Boundary, case, plural, and CJK semantics are identical; the replica's link counts are an inclusive upper bound within that frame. Because nesting prevalence can differ per lexicon, we re-ran all cross-domain arms under production-style overlap suppression (longest-match, non-overlapping): links fall 10–40% per arm but *every ordering conclusion is unchanged* (schema CS C_{10} 0.859 with identical entities; the size-matched arm C_{10} 0.323 vs. three-source 0.366; frequency still floods at 23+ concepts/doc). An earlier single run had sampled an ordered prefix—a bias we replace rather than defend; all metrics below are deterministic, reported as mean±sd across the ten resamples. Concepts-per-document is averaged over *matched* documents only (those with ≥1 scan hit under the arm's dictionary: mean 977 matched documents per sample under v2.0, 1,579 under v2.1), so it is not an all-sampled-document mean, and the two arms' denominators are different sets. On the full 2,000-document denominator (unmatched documents count zero) the same arms read 0.90 vs. 3.07 concepts per document (×3.4): the matched-subset ratio (×2.1) conditions on the documents the weak arm can annotate at all, and therefore understates the coverage gain—the selection effect runs against the collapsed dictionary.

Three findings. **(1) The collapse signature is reproduced by the lexicon alone**: 109±5 entities at concentration 0.513±0.016 and 1.85±0.05 concepts per *matched* document from the v2.0 dictionary—same

code, same corpus, no other variable. **(2) The repair effect is isolated**: v2.0→v2.1 multiplies link-bearing entities ×6.3 and concepts per matched document ×2.1 (the denominator grows too: mean link-bearing documents per 2,000-chunk sample rise 977→1,579), while concentration decreases 0.513→0.300 and generic-head mass falls 0.146→0.058; the production collapsed concentration (C_{10}=0.419 at the 10-06 backup) sits between the v2.0 (0.513) and v2.5 (0.322) counterfactuals, consistent with the operational dictionary having evolved mid-enrichment after the bulk of extraction had run. **(3) The "be" class is treatable at known cost**: case-sensitive short-symbol matching removes the single largest entity and takes concentration to 0.265±0.012, at the price of a wider case-fold recall loss (links −12.8%); a stop-word block is gentler (−8.1% links, C_{10} 0.252±0.011). Judged precision is near-tautological for dictionary self-matches (the entity name *is* the surface form), so the structural metrics lead and precision is a bounded secondary—the same caveat the link audit carries. These deltas measure matching *output*, not true recall: the removed mass mixes errors with correct terms, and the 0.580 pre-intervention judged-valid rate on the short-ambiguous-symbol class motivates the treatment—the removed-vs-retained verification below prices exactly this cut.

Budget sweep, closing the size confound. A natural concern is whether the isolated comparison (182-term collapsed vs. 3,733-term repair dictionary) confounds size with selection. We re-ran the frozen KB1 frame at fixed budgets $K \in \{200, 500, 1{,}000, 2{,}000, 3{,}733\}$ with three same-K sources: the v2.1 pipeline in construction order (prefix), frequency-top-K, and C-value-top-K (candidates ranked on sample-0 text; frame overlap disclosed). Coverage under the dual-annotated reference grows monotonically with v2.1's own budget (R 0.004→0.043 from K=200 to 3,733) while concentration falls 0.73→0.30 — the repair's value accrues with budget but its structure at every budget is distinct: frequency-top recruits 4–5× the entities (2,036 at K=2,000) for nominally higher R that rides on mass its own blind validity disqualifies (0.45 at the equal-size control), and C-value-top saturates at 554 entities with R stuck at 0.012 — statistical salience without attestation stalls on this corpus. Re-running the whole sweep with the two baselines re-typed by the unified classifier leaves R identical at every budget (0.001/0.016/0.037/0.073/0.098; entity counts within 1.5%)—the budget-sweep conclusions are typing-invariant. Size explains the counts; it does not explain the validity or the distribution.

Verification: semantics, governance, monitors. Three follow-ups

on the same frozen samples close the gaps declared above. *(i) Blind term-mention judging.* The replica folds matched surfaces onto themselves, so the production audit's does-the-span-denote-the-canonical question is tautological here; we ask instead whether the matched span is, in context, a genuine terminological mention of the term (ordinary-language use of the same letters fails). Entity-uniform, 150 links per arm, judge Qwen3.6-35B at temperature 0 (same family as the lexicon classifier—disclosed, and the same exposure the cross-model check bounds). Result: **old and new dictionaries are locally indistinguishable**—0.940 [0.890, 0.968] of v2.0's matched spans are genuine term mentions vs. 0.927 [0.873, 0.959] for v2.1 (domain share among valid: 89/141 vs 89/139). The collapse is therefore not per-link falsehood: category-word links are individually "valid" mentions of generic concepts. What fails is the vocabulary and its mass—109 entities at C_{10}=0.513 vs 692 at 0.300—and a per-link precision audit, however large, would not have caught the incident. *(ii) Equal-size frequency control.* A 3,733-term dictionary drawn purely by corpus frequency (same filters, no seed transplant, no LLM classification, no expansion; types randomly assigned from v2.1's inventory) reads **0.450 [0.356, 0.548]** on the same frozen text, and only 15 of its 45 valid links name domain concepts (v2.1: 89 of 139): the dictionary-size confound is answered directly—size buys nothing, selection carries the validity. Its structural profile is itself diagnostic: 1.70M links on the same 20k chunks (17× v2.1's), with a head of markup fragments and function words (`td`, `of`, `sup`, `mathrm`)—the frequency pipeline without the classification gate does not merely underperform, it re-collapses the link layer through a different door. *(iii) Governance verified.* Short-symbol hits that case-sensitivity removes (case-mismatched surfaces of ≤3-letter ASCII terms) are **0.033 [0.009, 0.114]** valid; exact-case retained ones are 0.700 [0.575, 0.801]: the −12.8% link cut removes almost purely error mass, and the retained class still carries 30%—the treatment is demonstrated post-hoc, not merely motivated. *(iv) Graded-degradation monitor validation.* Two axes on the ten frozen samples, no gold labels: a *replacement* axis mimicking the incident (drop m% of v2.1, keep all category words) and a *shrinkage* axis (random deletion). The C_{10} band fires at ≥75% replacement (0.371) and at 100% reproduces the collapse arm bitwise (0.513/109 entities—the replica's cross-script determinism check); count monitors fire earlier under shrinkage (links −50% at 50% deletion) where C_{10} stays flat to 50%; and moderate replacement (10–50%) moves C_{10} by less than one sd while counts drop up to 41%—a blind zone for both single-monitor families. Conclusion unchanged

from the design argument, now with evidence: gold-free monitors deploy as a set, and the name audit remains the sharpest early discriminator of the vocabulary failure that per-link and count signals both miss.

Short-symbol ambiguity: four linking policies. On 150 short-symbol mentions sampled from the frozen replay chunks (60 on the `be` class; each mention judged once, blind, by judge A and the cross-family judge, 12 disagreements), four policies on the same dictionary: linking everything (current behavior) reads precision **0.24**; exact-case gating (tested here, not yet deployed) 0.725 at true-symbol recall 0.806, rejecting 73% (rejection precision 0.936); adding context evidence (long-term neighbors in a ±150-character window) recalls slightly more (0.833) at lower precision (0.652); context plus reverse-NIL trades recall (0.583) for precision (0.724). On the `be` class specifically, *zero* of 60 sampled mentions are genuine element references—the exact-case gate removes pure error there. Context gating and case gating exchange precision for recall rather than dominating each other; the recoverable mass lives in the non-`be` short symbols, where identity (not term-ness) is the open problem.

Why 692 entities here vs 1,871 in production? The isolated frame counts only entities induced by scanning 2,000 chunks (0.034% of the library) under replica folding: rare terms need corpus mass to be attested, and the production layer additionally carries entities from the full corpus and historical merging. The causal claim rests on within-frame comparisons, never on this cross-frame count. Methodological caveats: semantic-replica scanner, 2,000-chunk samples (ten resamples bound the variance but not the coverage of a 5.83M-chunk library), and committed-snapshot arms rather than independently regenerated component ablations (the provenance decomposition in Section 7 approximates that axis lexically).

Provenance decomposition (component contribution, approximated lexically). The v2.1 file carries no per-term source tags, so we decompose its isolated link mass by which earlier committed lexicon first contained each matched term: v2.0-inherited terms contribute 22.0% of links (70 entities), mid-evolution terms 16.0% (232), and terms first introduced by the three-source repair **61.9% of links (384 of 686 entities)**—the repair's additions carry three-fifths of the rebuilt link mass (shares stable against the retired ordered-prefix frame, which read 21.4/15.3/63.2). A true component ablation (independently regenerating single-source dictionaries) remains future work; this set-algebra attribution is its lexical lower bound. A scope note on dictionary size: the arms vary the dictionary *as a whole* (v2.0 and

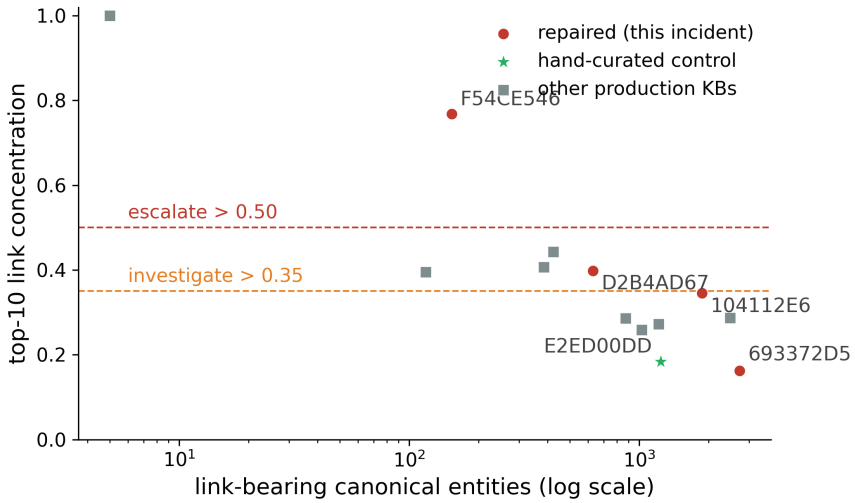


Figure 9: Top-10 link concentration vs. link-bearing entities across the 13 production libraries (log x-axis). Red = libraries rebuilt in this incident; green star = hand-curated reference case; gray = remaining production libraries. The bands are the proposed post-hoc alarm thresholds; the smallest libraries sit high partly by the Zipfian floor effect discussed in the text.

v2.1 differ by ∼20× in scan terms), so the contrast identifies the dictionary change, not its decomposition into term count, term-selection quality, and per-source method; the equal-size frequency control and the blind old/new matched-context judging in the verification below supply both controls on the same frozen samples (the public-domain cached-component ablation of Section 8 supplies the single-source analogue; independently regenerating single-source dictionaries for the *production* library remains future work).

7. Discussion

Observability for lexicon quality. Four cheap monitors would have caught both incidents at first symptom, in the spirit of production data-validation systems [11]:

(a) *Top-10 link concentration (primary).* The fraction of all entity links held by the ten most-linked entities—a concentration diagnostic invariant to uniform rescaling of link counts. That specific invariance is what

raw cardinality lacks; it is not a license to read the value as a corpus-size- or discipline-independent quality scale (topic concentration moves it too). Measured across the thirteen libraries of our deployment, the metric spans 0.16–0.77 among libraries with ≥100 entities (a five-entity library reads 1.000—the $\min(10, N)/N$ floor bound, not degradation; $N \leq 10$ forces C_{10}=1, and the metric also responds to topic concentration as well as to degradation): the healthy and repaired large libraries run at 0.16–0.19 (rebuilt physics KB2 0.162, hydrogeology control 0.184); the rebuilt nuclear library sits at 0.345 with its generic tail; and the two small treated libraries **currently** read 0.398 (1,324 docs, 629 entities) and 0.768 (119 docs, 153 entities)—values we report as they are, not smoothed: the smallest library also illustrates the metric's floor effect (with $O(10^2)$ entities, ten of them can hold most links under an ordinary Zipfian distribution alone), so the band should be read jointly with library size. The collapsed state itself is now measured, not gestured at: recovering the pre-incident cold database backup (2026-10-06, two days before discovery, with the collapsed enrichment still writing—4.8M links on KB1 at that moment versus 19.2M at discovery) gives **0.419** for collapsed KB1 (top-10 entities: *time, model, method, energy, temperature, field, term, tau, phase, material*—431k links on "time" alone) and **0.290** for collapsed KB2. The repair moves KB1 0.419→0.345 and KB2 0.290→0.162—direction correct, and the residual gap is the generic tail. Note also that collapsed KB2's 0.290 sits *below* the 0.35 investigation band: a concentration threshold alone does not catch every collapse—KB2's is exposed by the name-audit and coverage legs, which is why the monitors deploy as a set. Nor should KB2's repaired 0.162, *below* the hand-curated control's 0.184, be read as outperforming the control: a probable contributor is its thin repaired lexicon (v1.1, 1,279 terms, a declared floor), since vocabulary size reshapes the head share for reasons orthogonal to term quality—concentration is a distributional diagnostic, not a scalar quality scale. One honesty note: the metric partly improves simply because the entity population grows (503→1,871 link-bearing entities); the name audit (c) remains the sharper discriminator of the collapse itself. We propose a band alarm (>0.35 investigation, >0.5 escalation) with an explicit caveat: these are *empirical, post-hoc* candidate thresholds read on this deployment's thirteen libraries, not on an external population—triage triggers for human inspection, not classifiers. Two detection targets

must be kept distinct. *Collapse detection* asks whether the entity layer has degenerated; *lexicon-quality warning* asks whether the link distribution is contaminated with generic vocabulary. For collapse detection, a healthy library whose terminology is genuinely concentrated is a C_{10} false positive; for a lexicon-quality warning the same trigger is correct when the tail is in fact generic (the small-library case below). The composite rule's zero-false-alarm claim below is scoped to collapse detection on the graded-degradation archive. Their known blind spots, stated as such (per target). *False positives* (collapse target): healthy libraries whose terminology is genuinely concentrated (the two small treated libraries at 0.398/0.768—real terms, small corpora; the public-domain CS three-source arm at 0.404, Section 8). *False negatives*: replacement-type degradation below $\sim$75% substitution, where C_{10} moves less than one sd (offline graded degradation on the frozen resamples, Section 6; the experiment has no time axis, so we make no detection-latency claim), and collapses whose head happens to be diversified (the collapsed physics backup reads 0.290, below the band—count monitors and the name audit carry those cases). Moderate replacement is a blind zone for both signal families—monitors deploy as a set; libraries below a few hundred entities warrant size-stratified bands. A leave-one-library-out check on the 14 rows (thresholds re-frozen at the q90/q97 of the remaining healthy libraries each fold) yields 12 correct quiets, the same known false-positive class (the genuinely concentrated small library), and the same known false-negative class (the collapsed backup reads below the LOO band—count monitors and the name audit carry it): the bands are self-consistent across libraries but remain deployment-derived; unseen-deployment validation stays future work.

Composite rule, evaluated. Because single-leg thresholds misfire in both directions, the deployable unit is a four-leg disjunction (fire on any leg): concentration (C_{10}>0.35, small libraries exempt), link-count drop (>50% vs. healthy baseline), generic-head share (>0.10), and entity-cardinality drop (>50%). On the frozen graded-degradation archive this composite fires on every should-alarm state (collapse, $\geq$75% replacement, $\geq$50% shrinkage) with *zero* misses and zero false alarms on the healthy baseline and mild degradations (collapse target; the naturally concentrated small libraries are exempt from this leg, where the

same reading remains informative as a lexicon-quality warning). We report this as *in-sample diagnostic verification*, not external validation: thresholds, degradation classes, and pass criteria all derive from the same production libraries and frozen samples, so the result shows the rule reproduces the intended behavior, not that it generalizes to unseen knowledge bases; a frozen-threshold test on unseen libraries remains open. The 10–50% replacement band remains undetected by every leg (at 50% the generic-head leg sits at its threshold, 0.096 vs. 0.10) — the honest structural floor of aggregate-statistics monitoring, and the reason the name-audit leg exists. *Threshold sensitivity.* The composite's verdicts are stable to perturbing the generic-head threshold τ=0.10 by $\pm$0.02. Across $\tau \in [0.08, 0.12]$ the archive verdict is unchanged: every should-alarm state still fires and no healthy or mildly degraded state does—their highest generic-head reading is 0.073 (25% replacement or shrinkage), a margin of 0.027 below the nominal threshold, $\sim$7$\times$ the leg's resample sd (0.004). The 75%-replacement state clears three legs at once (C_{10} 0.371, generic head 0.104, links $-$58.6%), so its detection does not hinge on any single threshold. One boundary placement is nonetheless consequential: the 50%-replacement state reads 0.096—four thousandths under τ=0.10—so at the nominal threshold it is the archive's single blind-zone state, while at τ=0.09 the generic-head leg catches it with no new false alarm on any healthier state. In-sample, the blind zone's leading edge sits within measurement noise of the threshold; a deployment whose healthy head share is low enough may read the band down to 0.09 and trade that false-negative class for zero observed false positives on this archive, re-reading the choice against its own healthy distribution. *Cold start.* The two count legs (link-count and entity-cardinality drops) compare against a healthy history baseline that a newly created library does not have; until a first full enrichment pass completes and its post-pass snapshot seeds the baseline, those legs are inert by construction, and protection rests on the two history-free legs—top-link concentration and the top-entity name audit—which are exactly the legs that exposed this incident. The thirteen-library candidate calibration, summarized (the collapsed row is the pre-repair backup, not a live point):

Group	Libs	Entities	C_{10} range
Collapsed (10-06 backup)	1	503	0.419
Hand-curated control	1	1,239	0.184
Other production, ≥500 entities	4	874–2,482	0.259–0.287
Rebuilt here, large	2	1,871 / 2,726	0.162–0.345
Rebuilt here, small	2	153–629	0.398–0.768
Other production, <500 entities	4	5–424	0.395–1.000

Entity counts are monitoring-snapshot values (2026-10-12); the rebuild-completion counts (Section 5) read 2,730 for the same library — drift of four entities from post-rebuild enrichment, not an inconsistency.

(b) *Entity cardinality ratio (coarse secondary).* Canonical entities per ingested document, alarmed against a per-domain band (our control suggests O(10–20) for scholarly corpora; collapse ran at 0.009, a factor 2,192 below the reference case, and the ratio's sublinear numerator makes it scale-sensitive—the per-document concept counts above are the better-behaved metric). Cardinality alone is gameable in the other direction—the small-KB treatment showed category-word aliases *inflating* the count (Section 10)—which is why it backs up (a) rather than leading. The O(10–20) band itself comes from a single hand-evolved control library and is offered as illustration, not a calibrated norm; that control is textbook-scale material (∼1,913 chunks/document vs. 99 for KB1, with comparable per-chunk link rates of 4.8 vs. 4.1), so its per-document band transfers to journal-article corpora only as an order of magnitude.

(c) *Top-entity name audit.* The top-K entities by link count should be domain concepts; a manual or LLM spot check of that list is a five-minute quarterly control, and the link-precision rubric of Section 6 is a ready-made protocol for a deeper sample.

(d) *Target-table growth reconciliation.* Completed-job counts reconciled against downstream row growth, mandatory wherever a bulk writer is guarded by INSERT IGNORE (the ghost-bucket incident of Section 5—rows landed in an all-zero kb bucket invisible to every per-KB query while job metrics stayed green—was exposed by exactly this reconciliation).

None of the four requires ground truth. The small-library treatment (Section 10) is an illustrative within-deployment case for the monitor stack—the candidate bands were read *after* observing these thirteen libraries, so this is not out-of-sample validation: after the same repair playbook, entity counts contracted (870→629, 209→153—the cardinality signal normalizes) while the concentration monitor still flags both libraries—one above the investigation threshold ($0.398 > 0.35$), one above the escalation threshold ($0.768 > 0.5$)—correctly *as lexicon-quality warnings*, since their tails remain generic—demonstrating that the two monitors separate "cardinality repaired" from "distribution still skewed", which cardinality alone cannot.

What the repair restores, and what it does not. The evidence supports two claims of different strength. At the *entity-layer structural* level the repair succeeds on every deployed observable: entity diversity (575→1,871/2,726 canonical entities), annotated-document coverage (48.5%→98.5%), concentration back inside the healthy band, and link precision confirmed by stratified audit. At the *terminology-coverage* level the repair is deliberately partial: the from-scratch enumeration (Section 8) bounds coverage of human-recognized terms at 16.9% of core recurring constructions under the size cap, and half of the enumerated terms are corpus-unseen coinages that no bounded static lexicon can carry. Three-source construction is therefore a high-validity, bounded-lexicon-budget *repair* for the entity layer—not a high-recall general-purpose term-extraction method; low-frequency and newly coined terminology is mention-layer scope, where a bounded lexicon hands off to runtime extraction.

LLM extraction versus dictionary scan. The incident is not an argument for LLM-everywhere. It is an argument for honoring the input contract of cheap deterministic components: if extraction is a scan, the dictionary is the model—and it deserves the same evaluation discipline (held-out attestation, distributional checks) as a trained model. Batch-generating that dictionary with an LLM and shipping it on structural validation alone is the actual failure mode.

Schema tooling implications. The batch schema generator produced category taxonomy where a terminology was needed; the repair gave the schema a real lexicon (3,702 terms) and folded lexicon generation into the schema lifecycle as an idempotent tool. What remains open—and what the rebuilt top-entity distribution makes plain—is that *admission* criteria (corpus attestation) are necessary but not sufficient: a lexicon also needs *ranking* criteria that demote high-frequency generic words and short ambiguous symbols

(IDF-style corpus contrast, bilingual alias consistency, length/type priors). The observation that the collapsed state and the repaired state differ by a factor of 3.25 in entity cardinality—while both pass structural validation—is the strongest argument we have for treating lexicon quality as first-class schema infrastructure with its own acceptance gates, not as a one-time repair.

8. Does It Generalize? Three Public Domains

The incident corpus is nuclear-science literature; a natural objection is that the failure and the repair are local accidents. We therefore replay the lexicon-only design on three public domains—arXiv cs.CL (10,000 abstracts), PubMed biomedicine (9,988), and arXiv cond-mat.mtrl-sci (10,000)—with seeded splits per domain: 3,000 abstracts for lexicon construction only, 100 frozen test abstracts for silver annotation, and the remainder replayed as ten disjoint ∼690-document samples for structural metrics (mean across samples; sd archived). *Mass* throughout denotes two reported quantities—lexicon size (|L|) and per-document annotation volume (concepts/doc)—plus matched-document coverage where stated; we report all three per arm and predefine no scalar utility over them, so no "best" is asserted. Five arms share one scanner; only the lexicon varies: *batch schema* (the production-failure form: each domain's *own* current repository schema lexicon, i.e. the batch tool's committed output for that domain—category-shaped alias sets, in CS largely Chinese or uppercase-acronym forms), *frequency* top-K and *C-value* top-K (each size-matched to ours), *LLM direct* (per-batch extraction without corpus validation), and the *three-source* recipe. One adaptation is disclosed: source 1 transplants seeds from a healthy near-domain library, which does not exist for these public domains—we substitute LLM knowledge seeds under the same corpus-presence validation; sources 2–3 (corpus-candidate classification, per-type expansion with presence validation) are unchanged. The replay harness (scanner replica, sampling, judging protocol) reuses the companion system's tooling [17].

Model-annotation protocol with human verification. The annotation layer is model-based; human input enters as verification of a stratified sample (below), not as from-scratch labels. Silver reference lists come from a strong annotator LLM (judge A, Qwen3.6-35B) enumerating every domain term in each test abstract; arm quality is then judged blind on stratified samples (head/medium/tail/short-symbol entities) by judge A and a second Qwen-series model (judge B), with disagreements and missing judgments ar-

bitrated by a third *cross-family* judge (DeepSeek; 55–71 rows per domain; 8–29% of rows per cell). Two instrument limits are disclosed rather than hidden. First, enumeration is the weak link: on 30-document subsets re-annotated by the cross-family model, term-list Jaccard with judge A is 0.41 (CS), 0.55 (biomedicine), 0.52 (materials)—so silver P/R are indicative levels, not ground truth, and we read contrasts between arms (which share the instrument) rather than absolute magnitudes. Second, judge B cannot enumerate at all (the gateway forces chain-of-thought past read timeouts) and is used only for short blind judgments; and the LLM-direct arm's silver-P is same-family inflated, since judge A's family also built that arm.

Instrument sensitivity, tested on the shared subset. Because a shared instrument does not guarantee arm-wise bias cancellation, we recomputed all five arms' occurrence-level P/R on the exact 30-document A∩C subset under *both* references (matching rule: lower-cased surface with naive plural stripping, applied symmetrically). The recall ordering of the five arms is **identical under both judges in all three domains**; the F1 ordering is identical in biomedicine and materials, while in CS only the middle band permutes (frequency/C-value/three-source swap ranks 2–4; first and last unchanged). Three-source's precision *rises* under the cross-family reference in all domains (CS 0.218→0.312, biomedicine 0.195→0.287, materials 0.246→0.260) while LLM-direct's falls (CS 0.653→0.544), consistent with the disclosed family bias and mildly favoring three-source under the harder reference. Cross-domain conclusions therefore rest on (i) structural metrics (no instrument), (ii) recall ordering, and (iii) arbitrated blind validity; mid-band F1 orderings and absolute P/R levels are instrument-conditional. Judged-valid rates carry Wilson 95% CIs—e.g. biomedicine batch-schema 0.906 [0.810, 0.956] vs. three-source 0.850 [0.756, 0.912]: overlapping. *A dual-annotated reference, and what survives it.* To strengthen the instrument we finally cross-annotated 200 complete units — 80 frozen KB1 chunks plus 40 test documents per public domain — with two independent model annotators reading full text (not approving candidates): span-level entries with termness, specificity, type, and identity-ambiguity flags. P1 = deepseek-flash, P2 = deepseek-v4-pro (same vendor, different models — overlap disclosed; a gateway thinking-mode model was attempted first and is unusable for full-document annotation: forced deliberation exceeds any practical token budget), disagreements arbitrated by the cross-family Qwen judge. Agreement is materially stronger than single-model enumeration: domain-set Jaccard **0.59–0.66** across the four scopes, termness agreement on overlap

94.5–98.8% (vs. 0.41–0.55 for enumeration pairs). Under this reference: the batch-schema arms recall 0–1% (the collapse signature under the strongest instrument yet); the production lexicons' first recall estimate on KB1 reads v2.0 1.0% → v2.1 4.3% of adjudicated distinct concepts (4.3×; the reference is deliberately exhaustive, so absolute levels are floors, not ceilings); the selection arms' recall deficit replicates (learned/cover 0.7–3.2%); and the frequency arm's apparent recall rides on mass its own blind validity (0.29–0.40) already disqualifies. *Human verification*: one author independently confirmed a seeded, stratified sample of 197 adjudicated terms over 20 units (5 per scope; sheet, protocol, and per-row verdicts in the artifact) — confirm rate **0.934** [0.890, 0.961]; by scope 0.833 (KB1) / 0.967 / 0.925 / 0.950; two-model consensus items verify at 0.949 vs. 0.853–0.960 for arbitrated additions. This bounds the reference's precision from the model side; it does not measure omissions (verification, not from-scratch re-annotation) and is single-author — an independent two-annotator gold remains future work. A third gateway model (Qwen family, short-task capable) independently confirms the shared domain-term decisions at 0.75 (KB1) and 0.875/0.875/0.95 (CS/biomedicine/materials) on 40-item samples per scope. Entity-cluster bootstrap CIs for the three-source validity rates (10,000 resamples) are [0.747, 0.921] (CS), [0.772, 0.924] (biomedicine), [0.846, 0.974] (materials) — wider than link-level Wilson intervals, conclusions unchanged.

From-scratch enumeration (recall side): to bound the omission side that verification cannot see, the same author enumerated domain terms from zero on 30 additional held-out abstracts (10 per domain, sampled from the frozen cross-domain test splits; no candidates were shown), yielding 418 unique terms matched against the five arms by exact surface after case folding (Table 6). The batch-schema arm covers **0/418** — including 0 of the 83 terms attested in more than ten construction-corpus documents: the collapse signature replicates under the strongest instrument yet, now from the recall side. Half of the enumerated terms (209/418) never occur in the 3,000-document construction corpus — paper-specific coinages (new method, benchmark, and model names) that no bounded static lexicon can carry and that belong to the mention layer; this bounds what lexicon-layer recall can mean at all. Within the core stratum the size-aligned arms cover 16.9–22.9% (frequency 14/83, C-value 19/83, three-source 14/83) and the unvalidated LLM-direct arm 47.0% with a 4.6× larger lexicon; per 1,000 lexicon entries the recall-efficiency ordering is C-value 16.6 > three-source 14.8 > frequency 12.2 > LLM-direct 10.5. The probe prices the validity -first selection honestly: under the shared size

Table 6: Recall-side reference: from-scratch human enumeration (30 held-out abstracts, 10 per domain; 418 unique terms) matched against the five cross-domain arms by exact surface (case-folded). Strata are construction-corpus document frequency (3,000 docs per domain). The LLM-direct lexicon is 4.6× larger on average and unvalidated (silver-family bias, indicative).

Stratum	n	schema	freq	C-val	llmdir	3-src
unseen (df=0)	209	0	0	0	0	0
rare (df=1–2)	59	0	0	0	4	0
recurring (df=3–10)	67	0	0	0	12	3
core (df>10)	83	0	14	19	39	14
all terms	418	0	14	19	55	17

cap, three-source trades a modest recall deficit to C-value for its arbitrated blind validity (0.838–0.912 vs. 0.27–0.40—model-judged orderings throughout; an independent human gold is the declared next step). The enumeration sheet (git-ignored: public-corpus text), per-term document-frequency counts, and the strata table ship with the artifact.

The collapse signature travels (Table 7, Figure 10). The batch-schema arm on CS is the incident in miniature: the domain's *own* committed schema lexicon yields C_{10}=0.899 with blind judged-validity 0.565 and 18 matched entities; only 5.7% of documents carry any annotation (its aliases are category-shaped and largely Chinese or uppercase-acronym forms, absent from English abstracts) and silver recall is 0.000. Biomedicine and materials schema arms are milder—some of their category words ("disease", "cell", "phase") happen to be usable domain terms there, which is why their blind validity runs higher (0.688–0.906) while they still concentrate (C_{10} 0.43–0.83) at recall $\leq$0.009: concentration, not per-link validity, is the collapse signature. Frequency floods at any size: 25–28 concepts per document with validity 0.36–0.45—volume without terminology. C-value at equal size repairs concentration (0.26–0.55) but not validity (0.33–0.40): in abstract-length corpora, statistical salience is not termhood. Because these two baselines' entity types were originally assigned at random from the schema type list, the table reports both arms under *unified typing* (term sets frozen; types re-assigned by the same LLM classifier, model and prompt, that the three-source arm's classification stage uses; classifier-rejected terms stay as "unclassified" to preserve the fixed term set; random-typed rows archived in

Table 7: Cross-domain replay: five lexicon arms on identical frozen samples (structural metrics are means over ten disjoint ∼690-document samples; |lex| size-matched for freq/cvalue/threesource). Valid = blind judged-valid rate after cross-family arbitration (n=60–80 per cell); P/R/F1 are occurrence-level against silver enumerations—levels indicative only (see text). †Frequency/C-value rows report the *unified-typing* re-run (term sets frozen; entity types re-assigned by the same LLM classifier that three-source's classification stage uses); silver P/R/F1 re-run mechanically against the same silver references under the unified-typing term sets; the original random-typed rows are archived in Table B.9.

Domain	Arm	Terms	Entities	C_{10}	Conc/doc	Valid	P_s	R_s	F_1
CS	batch schema	95	18.4	**0.899**	1.14	0.565	0.000	0.000	0.000
CS	frequency K=251 †	251	286.0	0.319	24.8	0.405	0.091	0.097	0.094
CS	C-value K=251 †	251	255.0	0.554	5.1	0.400	0.299	0.043	0.075
CS	LLM direct	1198	344.5	0.501	4.6	0.787	0.663	0.132	0.220
CS	three-source	251	202.1	0.404	3.6	**0.838**	0.274	0.050	0.085
Biomed	batch schema	130	100.5	0.428	2.8	0.906	0.089	0.009	0.016
Biomed	frequency K=513 †	513	574.0	0.166	25.1	0.362	0.038	0.038	0.038
Biomed	C-value K=513 †	513	449.0	0.291	6.5	0.400	0.072	0.022	0.034
Biomed	LLM direct	2003	591.5	0.309	4.9	0.838	0.343	0.085	0.136
Biomed	three-source	513	408.7	0.281	5.1	**0.850**	0.273	0.060	0.098
Materials	batch schema	152	58.5	**0.829**	2.3	0.688	0.054	0.004	0.007
Materials	frequency K=384 †	384	448.0	0.151	28.2	0.450	0.028	0.035	0.031
Materials	C-value K=384 †	384	390.0	0.256	6.8	0.333	0.153	0.044	0.068
Materials	LLM direct	2045	713.8	0.198	6.9	**0.925**	0.336	0.093	0.146
Materials	three-source	384	337.4	0.268	5.4	0.912	0.223	0.046	0.076

Table B.9). Random assignment had disagreed with the classifier on 91–98% of classifiable terms; unified typing shifts the two baselines' validity by −3 to +13 points (frequency 0.405/0.362/0.450 and C-value 0.400/0.400/0.333 in CS/biomedicine/ materials, vs. 0.293–0.400 and 0.267–0.367 under random typing; all within overlapping intervals at n=60–80). Typing was therefore a bounded confound, and correcting it nowhere closes the gap to three-source (0.838/0.850/0.912): the dominant deficit is candidate selection. The same classifier accepts only 6–19% of the frequency arm's terms—running frequency through the three-source gate would retain 36 of 251 CS terms, which is the selection mechanism speaking, not the type labels. LLM-direct

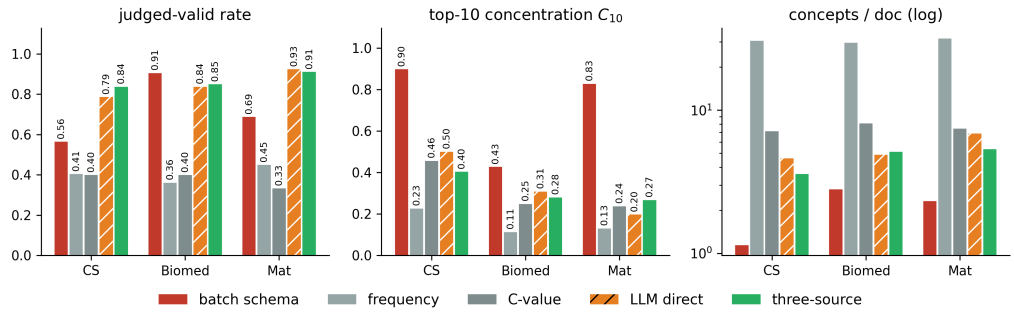


Figure 10: Cross-domain replay at a glance. Left: blind judged-valid rate after cross-family arbitration. Middle: top-10 concentration (the collapse signature; note the schema arm). Right: concepts per document on a log scale (the flooding signature; note the frequency arm). The LLM-direct arm is hatched where its silver-P reads same-family inflated. In the left panel the frequency/C-value bars show the unified-typing re-run (types re-assigned by the shared LLM classifier); random-typed rates archived in Table B.9.

reaches the highest validity in materials (0.925 [0.846, 0.965]) and the highest silver-P everywhere—but at 4.7–8× the lexicon size with no presence control, and its silver-P is the same-family-inflated column. Among the three *size-matched* arms, three-source holds the best arbitrated validity in CS (0.838 [0.742, 0.903]) and biomedicine (0.850 [0.756, 0.912]) and sits within overlapping intervals of LLM-direct in materials (0.912 [0.830, 0.957] vs. 0.925) and of batch-schema in biomedicine (0.850 vs. 0.906 [0.810, 0.956]—whose recall is 0.009, so its validity lead does not qualify it as a lexicon); its recall is at or above the frequency baseline in two of three domains at one-half the C-value arm's and one-eighth the frequency arm's annotation volume per document. Generic-head share is 0.5% (CS), 3.4% (biomedicine), 7.0% (materials)—reported per domain because the head list is the incident corpus's category vocabulary: its biomedicine/materials drivers ("disease", "drug", "phase", "cell") are legitimate domain terms in those domains, so the cross-domain residual is substantially list–domain mismatch rather than pure genericity (the frequency arm reads 4.7% in CS, where the same words are generic). No single arm dominates all reported metrics; we present three-source as the design point that pairs size-matched validity with bounded annotation volume under this protocol, not as a uniquely optimal balance.

Language confound, resolved by control. Because the CS batch-schema lexicon carries Chinese and uppercase-acronym aliases against English abstracts, its failure could be read as language mismatch rather than semantic

shape. A language-aligned control (English-normalized aliases only: CJK dropped, ASCII lower-cased, 80 terms) reads C_{10} 0.859 with the same 18.4 entities and the same 5.7% matched documents as the raw arm — the Chinese aliases never matched English text, and the English ones that do are category words. Normalization does move one observable: raw 94.1 links fall to 65.9 (-30%), because case-variant aliases ("Algorithm" / "algorithm") compile to overlapping case-insensitive patterns that each counted the same span; folding duplicates deflates top-head counts unevenly, which is the whole of the C_{10} 0.899→0.859 shift. A translation control closes the remaining path: the schema's non-ASCII aliases are exactly sixteen type-name translations (the Chinese for "algorithm", "data structure", etc.), not domain-term aliases; translating them to English (LLM) adds two net-new surfaces that match the corpus zero times, and every metric is identical to the deletion-based alignment. Deleting versus translating the Chinese content is a distinction without a difference—there is no Chinese terminology to lose. The CS collapse is semantic, not orthographic.

What carries the recipe (CS ablation, same frozen samples; Table 8). Removing corpus-candidate classification collapses coverage (concepts/doc 3.60→1.63): classification is the coverage workhorse. Removing presence validation keeps 977 raw entries for almost the same coverage (3.92 vs 3.60)—the gate removes 74% of raw LLM mass at ∼8% coverage cost, which is precisely the trade the incident demanded. Seed and expansion alone are thin (1.3–1.5 concepts/doc); unvalidated knowledge seeds carry 13× the entries of validated ones (768 vs 58) for barely more matched entities (80.8 vs 50.9)—mass without coverage. The size-matched frequency control floods (30.7 concepts/doc, 4.5% generic head). The recipe's value is in the assembly and the gate, not in any single source. One component is demoted on this evidence: *expansion*. Removing it costs nothing structurally (entities 211.7 vs. 202.1, concepts/doc 3.73 vs. 3.60), and its standalone hits read 0.433 valid under double judging with only 0.50 judge agreement (n=60) — contested quality. We reframe expansion as an optional diversity module, not a necessary source; the load-bearing pair is corpus-attested classification plus the presence gate.

8.1. Where Recall Is Lost, and What Selection Can Buy

The silver recalls above are occurrence-level against an enumeration instrument; before optimizing selection we decompose where the loss sits. For

every silver occurrence we test four nested events: expressible by *some* candidate (C: a wide pool—all construct n-grams at frequency $\geq$2 plus definition-evidenced abbreviations plus all LLM candidate lists; 12,149/15,032/13,121 terms per domain), kept in the arm's dictionary (K), detected by the actual scan at that occurrence (D), and linked under normalization (L). The funnel is unambiguous: $P(C)$ is 0.44/0.45/0.41 (CS/biomed/materials)—*the candidate pool, not the ranking, is the binding constraint*; three-source keeps $P(K|C)$=0.15/0.21/0.14; scan and link are nearly lossless ($P(D|K \cdot C)$ 0.89–0.98, $P(L) \approx 1$). The pool gaps are two-character element symbols (excluded by the tokenizer's 3-character minimum), terms attested only in test abstracts (construct-side presence cannot see them), and neologisms; the selection gaps include, e.g., *LLM* and *large language model* themselves in CS.

Risk-constrained selection, tested and not adopted. A learned-rank/set-selection alternative at equal budget (linear feature scorer on corpus contrast, C-value, phrase structure, abbreviation evidence, and ambiguity flags, trained on construct-side silver; set-selection over construct mention structure; greedy-plus-swap within 1.5% of a beam-search reference) raises silver-P while recall falls outside CS, and arbitrated blind validity collapses (learned-rank 0.29/0.46 in CS/biomedicine vs. 0.838/0.850 for three-source)—the silver-P gain was an instrument artifact, and construct-side learned selection without attestation gates reproduces the invalid-mass regime the incident taught. The full account, including the budget-sweep and concentration evidence, is archived in Section Appendix A; the headline: within an attested pool and construct-side proxies, neither learned ranking nor coverage-linked selection dominates the deployed pipeline, and selection pressure alone manufactures concentration.

9. Conclusion

Three questions framed this study. *When does a structurally valid lexicon produce semantic collapse?* When its entries are category words: the scan-and-link machinery executes faithfully, and a single-variable dictionary swap reproduces the collapse signature—no pipeline bug required. *What does corpus-presence gating repair, and what does it leave?* It restores terminology and coverage (annotated documents $\sim$51.5%$\rightarrow$98.5%; link-bearing entities $\times$6.3 in the isolated frame) but admits a generic tail—presence is necessary, not sufficient, and ranking criteria are the open problem. *What can gold-free diagnostics see?* Distribution and names—the collapse signature—

Table 8: Source ablation on CS (identical frozen samples; terms column makes the size confound explicit). −presence = all raw components unvalidated.

Arm	Terms	Entities	C_{10}	Conc/doc	Gen-head
full three-source	251	202.1	0.404	3.60	0.005
−seed	221	194.1	0.437	3.35	0.005
−classify	93	70.2	0.601	1.63	0.000
−expand	232	211.7	0.397	3.73	0.004
−presence (raw)	977	260.9	0.374	3.92	0.004
seed-only (valid.)	58	50.9	0.666	1.52	0.000
seed-only (raw)	768	80.8	0.616	1.57	0.000
classify-only	181	169.0	0.463	3.18	0.005
expand-only	42	27.3	0.838	1.29	0.000
freq size-matched	251	312.4	0.227	30.7	0.045

but not per-link truth: blind judging shows old-dictionary links are locally as "valid" as new ones (0.94 vs 0.93), so the collapse hides from per-link audits; the semantic gain runs through vocabulary and mass (equal-size frequency control 0.45 vs 0.93), graded degradation bounds when the alarms fire (band at $\geq$75% replacement; counts earlier under shrinkage), and the judged rates themselves (0.84 overall, 0.58 short symbols, removed 0.03 vs retained 0.70) remain model-assisted readings with disclosed family exposure; an independent dual-annotator human gold set is the declared next evaluation step. The production incident supplied the motivation and the validation environment; the controlled lexicon swap, the stratified audit, and a three-domain public-corpus replay carry the evidence. A fourth question closes the loop: *does it generalize?* On three public domains the batch-schema form reproduces the collapse signature (C_{10} up to 0.899 with recall $\leq$0.009), frequency floods, and the three-source recipe is the size-matched design point with the best arbitrated validity at bounded annotation volume—a trade-off under the tested metrics and AI instrument, not a claimed unique optimum; the failure mode and the repair are properties of the construction method, not of one corpus.

10. Limitations and honesty

1. **The generic tail survives the repair**: top-linked entities remain short generic symbols; admission gates without ranking gates move the problem, they do not remove it.

2. **Retrieval attribution is bounded, not eliminated**: the channel ablation of Section 6 shows the trajectory does not run through entity-channel fusion votes, but the full decomposition (lexicon old/new × retrieval configuration on/off) has not been run—the graded-degradation verification leg (frozen slices, Section 6) is a first step; the full lexicon×configuration grid remains unrun.
3. **Coverage**: two knowledge bases rebuilt (61,137 + 13,408 documents) and, after the two main libraries drained (2026-10-09), the two smaller affected libraries (1,324 + 119 documents) with the same playbook—where entity cardinality *contracted* (870→629, 209→153, with 916k/26k links and 4.6k new typed relations): the category-word aliases had inflated the count, so cardinality is not the quality metric, name distribution is. The bilingual alias layer showed isolated anomalies (an entity whose mentions vanished between extraction generations) that we document but do not fully explain.
4. **Incident-response cost**: six secondary incidents across the repair window (one INSERT-IGNORE ghost-bucket mis-write whose logged duration was corrected by row-level timestamps, three half-open-connection wedges, a watchdog that mistook completion for failure and deleted the freshly rebuilt graph, one missing-index class) each extended the repair; we report them as findings, not hidden costs.
5. **No human-adjudicated ground truth** for entity correctness: the link-precision audit is LLM-assisted (with programmatic span verification); the stratified 300-link sample replaces the uniform 100 but entity-uniform sampling intentionally over-weights the tail relative to link frequency, and the cross-judge is same-vendor-family; the blind term-mention judging (term-mention rubric), the equal-size control verdict, and the removed-vs-retained governance verdicts (0.033/0.700) inherit the same family exposure and a rubric not directly comparable to the production link audit. A human-adjudicated stratified sample is the natural next evaluation step; a 100-item dual-source annotation sheet (production links plus cross-domain judged items, blind to model verdicts) ships with the artifact for exactly this purpose.
6. **The cross-domain study is model-annotated, with a single-author human verification of a 197-term sample that bounds precision, plus a single-author from-scratch enumeration of 30 held-out abstracts that references recall** (Section 8): silver enumeration agreement across families is 0.41–0.55; an instrument-

sensitivity rerun on the shared 30-document subset shows the arms' recall ordering stable under both references (F1 mid-band orderings in CS are instrument-conditional); one gateway judge is unusable for enumeration, and the LLM-direct arm's silver-P is same-family inflated; cross-domain conclusions therefore rest on structural metrics, recall orderings, and arbitrated blind validity, and the source-substitution adaptation (LLM knowledge seeds in place of seed transplant) is disclosed rather than hidden. The recall funnel and the selection study (Section 8.1) inherit the silver instrument for their loss attribution and training labels; the selection arms' dev labels come from the cross-family judge, and the negative result is stated under those proxies.

This appendix is archival reference material; it is not required for the main claims and ships as online supplementary material in the artifact release when the compressed version is submitted.

Appendix A. The selection study: tested and not adopted

Moved from the main text (third-review slimming); content unchanged.

Risk-constrained selection, tested and not adopted. The natural next question is whether a learned or set-selection objective beats per-term ranking at equal budget. We built the wide pool above, a linear feature scorer (corpus contrast, C-value, phrase structure, abbreviation evidence, short-symbol and generic-word ambiguity flags) trained on *construct-side* silver enumerated by the cross-family judge (60 documents per domain; test documents never touch pool, training, or tuning), and the set-selection objective over construct mention structure with proxy link-correctness and specificity scores (explicitly proxies: no calibration claims). Greedy-plus-swap local search reaches within 1.5% of a beam-search reference on the CS top-300 slice. At the same K as three-source the selection arms fail, and the manner of failure is itself a finding. Silver-P *rises* (CS 0.274→0.362, biomedicine 0.273→0.352, materials 0.223→0.526) while recall falls outside CS (biomedicine 0.060→0.015, materials 0.046→0.009; CS flat-to-up 0.050→0.053); doubling or halving the budget moves recall by less than half a point while precision decays — the pool ceiling of the funnel again; and the selected lexicons *concentrate* (top-10 concentration 0.69–0.73 in CS vs. 0.404 for three-source). The arbitrated blind judging then reverses the apparent precision gain: validity collapses in CS and biomedicine

Table B.9: Random-typed rows archived from the original cross-domain replay (same frozen samples and arbitration protocol as Table 7).

Domain	Arm	Entities	C_{10}	Conc/doc	Valid	$P_s/R_s/F_1$
CS	frequency	312.4	0.227	30.7	0.293	.101/.097/.099
CS	C-value	273.4	0.457	7.2	0.267	.300/.043/.075
Biomed	frequency	642.7	0.114	29.7	0.362	.038/.039/.038
Biomed	C-value	495.4	0.249	8.1	0.350	.073/.022/.034
Materials	frequency	481.4	0.131	31.8	0.400	.029/.035/.032
Materials	C-value	411.2	0.238	7.5	0.367	.155/.044/.069

(learned-rank 0.29 [0.19, 0.41] and 0.46 [0.32, 0.55], vs. 0.838/0.850 for three-source) and stays below three-source in materials (0.86–0.89 vs. 0.912, at one-fifth the recall). The silver-P gain was an instrument artifact — enumeration under-covers, which flatters arms whose extra mass is invalid — and construct-side learned selection without attestation gates reproduces precisely the invalid-mass regime the incident taught. We therefore keep the three-source pipeline as the repair baseline and report this as a bounded negative result: within an attested pool and construct-side proxies, neither learned ranking nor coverage-linked selection dominates it, and selection pressure alone manufactures concentration.

Appendix B. Random-typed baseline rows (pre-unification archive)

The original cross-domain table rows for the two statistical baselines, whose entity types were assigned at random from the schema type list; superseded in the main table by the unified-typing re-run (Table 7 note, silver columns included). Kept as the pre-unification snapshot of both baselines.

Appendix C. Secondary engineering incidents during the rebuild

Reported as findings, not evidence for the paper's claims; each is fixed and committed.

- **B90 — unbounded projection join.** Graph projection originally joined each document against *whole-corpus* entities and the *whole-corpus* link table; at 80k documents and millions of links a single projec-

tion took 25+ minutes, and 80 concurrent consumers saturated a 112-core MySQL (107 cores, load 500). Scoping the join to the document's own mentions (an index-ordered path) is semantically equivalent—an entity projects through documents that mention it—and took projection to seconds, database to 29 cores.

- **B91/B93 — link-insertion deadlocks.** Two deadlock classes in link insertion: unordered INSERT-SELECT acquiring gap locks in interleaved order (fixed by pinning insertion order to `mention_id`), and per-action round-trip inserts (fixed by folding all actions into a single UNION-ALL derived-table INSERT-SELECT). Deadlock error 1213 went to zero.

- **B94 — the $O(N^2)$ the new lexicon amplified.** Alias canonicalization checked every acronym pair quadratic in alias count—95 of 101 seconds per links task, burning the GIL while 16 threads queued. Caching canonical forms by schema signature (lexicon version + alias count) took it to 0.05 s. Net link throughput: **8/min → 1,100/min (137×)**—the collapse-era lexicon was so small it hid this bug for months; the *repair* exposed it in an afternoon. The 137× is a window-level aggregate over concurrent fixes (the deadlock repairs land in the same window); the cache's isolated contribution is measured at task level—95 of 101 seconds per links task—and the cache key (schema version + alias count) can in principle alias two same-size, same-version, different-content lexicon edits; version-pinned bumps (v2.0→v2.1) are the mitigation.

The last point deserves emphasis in prose, not just as a bullet: the collapsed lexicon (575 de-facto terms) masked an $O(N^2)$ hot path because N was tiny. Correcting the lexicon did not merely improve extraction—it changed the computational regime of the pipeline and surfaced latent engineering debt. Lexicon quality is load-bearing in the most literal sense.

Data availability

The lexicon-generation tool (three-source, corpus-attested), the rebuild driver, the link-precision audit driver with all 400 adjudications (the uniform 100 plus the stratified 300—spans, verdicts, reasons), the isolated-comparison and resample drivers with their sampled chunk-id

manifests, the scanner-parity driver and its 200-chunk frozen result, the cross-domain replay package (public-corpus download and five-arm drivers, silver/judge/arbitration records, arm lexicons, per-domain tables, and the silver-stability rerun), the channel-ablation runner, and the observability queries (top-10 link concentration, cardinality ratio, top-name audit, row-growth reconciliation) are released in a public artifact mirror at `https://github.com/simpleqt/knowledgenexus-artifact` (no access request needed; the institutional GitLab carries the full development history). Each released result directory carries a SHA-256 manifest of its JSON records so external reruns can verify snapshot identity. Public source corpora are not redistributed and are identified by doc-id manifests in the same package. For double-blind venues an anonymized mirror will be provided with the submission. Schema lexicons (nuclear_electronics v2.1, physics v1.1) ship as versioned assets in the same repositories. The journal corpus is third-party copyrighted; entity-layer aggregate statistics and name samples sufficient to reproduce the forensic analysis are tabulated in the released evidence pack.

Conflict of interest

The authors declare no competing interests.

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